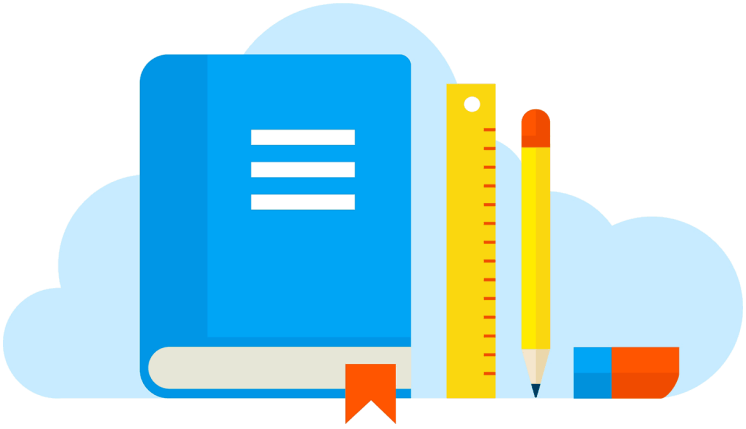


📘 NOTE 6 — NORMALIZATION IN DBMS

Date: 19/04/2026

1. Introduction

Normalization is a process of organizing data in a database to reduce **data redundancy** and improve **data consistency**.

In a poorly designed database, the same information may be stored multiple times. This can cause problems when data is inserted, updated, or deleted.

Normalization divides large tables into smaller, related tables.

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2. Need for Normalization

Normalization is used to:

- Reduce duplicate data

- Improve data consistency
- Organize database tables
- Avoid data modification problems
- Improve data integrity
- Make relationships between data clear

Here, the student's name and teacher information may be repeated.

Normalization can divide this information into separate tables.

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3. Data Redundancy

Data redundancy means storing the same data unnecessarily in multiple places.

For example:

- Student_ID: 101

  Name: Aarav

If the student's name is stored in many rows, the same information is repeated.

Problems caused by redundancy

- More storage is required
- Updating data becomes difficult
- Inconsistent data may occur
- Database maintenance becomes harder

Normalization helps reduce these problems.

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4. Normal Forms

Normalization is performed in different stages called **Normal Forms**.

The commonly discussed normal forms are:

1. First Normal Form — **1NF**
2. Second Normal Form — **2NF**
3. Third Normal Form — **3NF**
4. Boyce-Codd Normal Form — **BCNF**

Each normal form applies specific rules to improve database design.

Now each field contains a single value.

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6. Second Normal Form — 2NF

A table is in **Second Normal Form (2NF)** when:

1. It is already in 1NF.
2. There is no **partial dependency**.

Partial dependency occurs when a non-key attribute depends only on part of a composite primary key.

Simple idea

If the primary key consists of two columns:

Student_ID + Subject_ID

Then every non-key field should depend on the **complete key**, not just one part.

2NF is mainly important when a table has a **composite primary key**.

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7. Third Normal Form — 3NF

A table is in **Third Normal Form (3NF)** when:

1. It is already in 2NF.
2. There is no **transitive dependency**.

A transitive dependency occurs when a non-key attribute depends on another non-key attribute.

Example

- Student_ID → Department_ID

Department_ID → Department_Name

Here, `Department_Name` indirectly depends on `Student_ID`.

To improve the design, department information can be stored in a separate table.

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8. BCNF

BCNF stands for Boyce-Codd Normal Form.

BCNF is a stronger version of 3NF.

A table is in BCNF when every determinant is a **candidate key**.

BCNF is used when certain dependency situations are not completely handled by 3NF.

For basic database design, **1NF, 2NF, and 3NF** are commonly used.

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9. Advantages of Normalization

Normalization provides several benefits:

Reduced Redundancy

Duplicate data is minimized.

Better Consistency

The same information is less likely to have conflicting values.

Easier Updates

Data can be updated in one appropriate location.

Better Organization

Information is divided into logical tables.

Improved Data Integrity

Relationships and constraints can be managed more effectively.

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10. Normalization in StudySphere

Normalization can be applied to the StudySphere database.

Instead of keeping all information in one large table, different entities can be separated.

For example:

- Users
- │
- ├── Notes
- │

- ├── Assignments
- │
- └── Submissions

Users Table

Stores:

- User ID
- Name
- Email
- Password
- Role

Notes Table

Stores:

- Note ID
- Title
- Subject
- Description
- File information

Assignments Table

Stores:

- Assignment ID
- Title
- Subject
- Instructions
- Due date

Submissions Table

Stores:

- Submission ID
- Student ID
- Assignment ID
- File information
- Grade

This separation reduces unnecessary duplication and makes the database easier to manage.

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11. Normalization vs Denormalization

Normalization divides data into related tables to reduce redundancy.

Denormalization intentionally combines some data to improve read performance or simplify certain queries.

Normalization

- Students
- Assignments

Submissions

Data is separated into logical tables.

Denormalization

Some related information may be stored together to reduce the number of database operations.

Normalization is generally preferred when **data consistency and database organization** are the main priorities.

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12. Conclusion

Normalization is an important database design technique. It organizes data into related tables and reduces unnecessary duplication.

The main stages are:

1NF → 2NF → 3NF → BCNF

For applications such as StudySphere, proper database design helps manage students, notes, assignments, submissions, and grades efficiently.

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